

Workshop 2: Automated Data Analysis

The relationship between life expectancy vs GDP per capita for 2020 was analysed using automated data analysis. The results are presented below:

Fig. 1: Life Expectancy by country, 2020

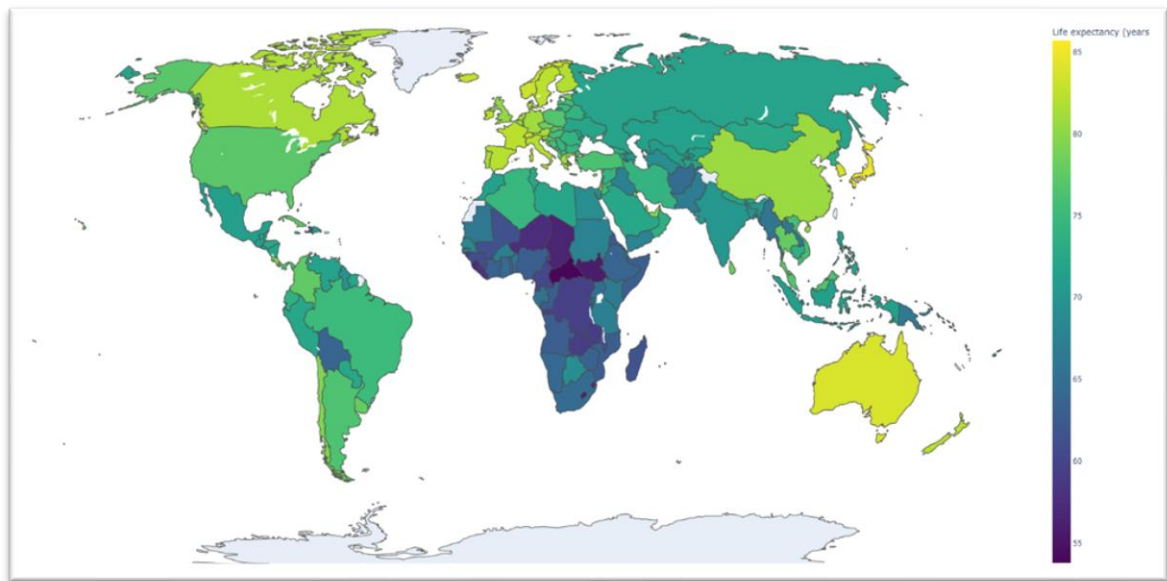


Fig. 2 GDP per capita by country, 2020

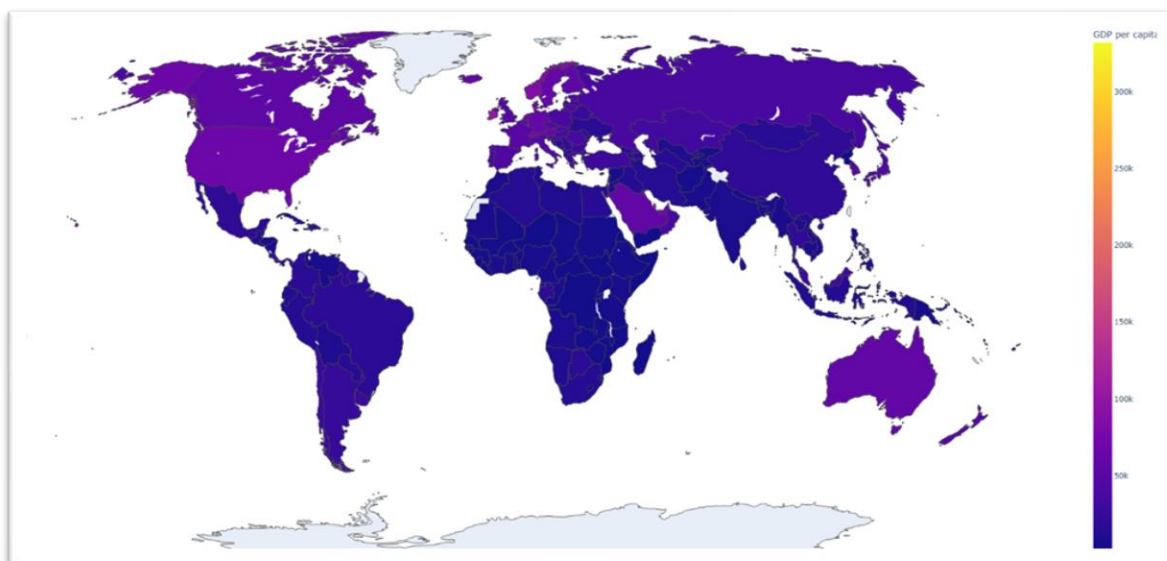


Fig 3: GDP per capita by country, 2020 shown in categories

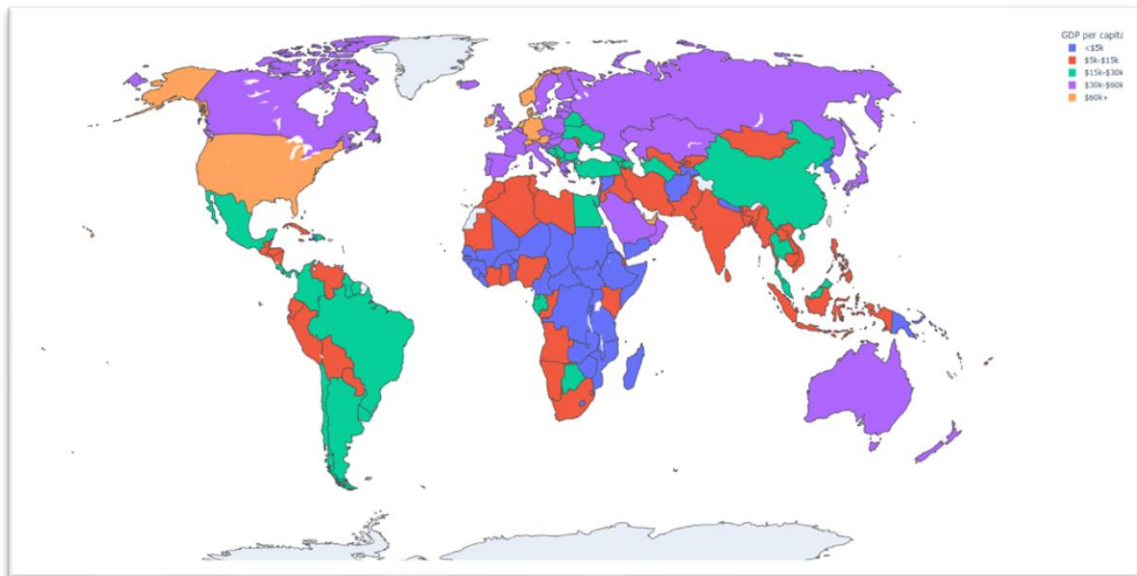
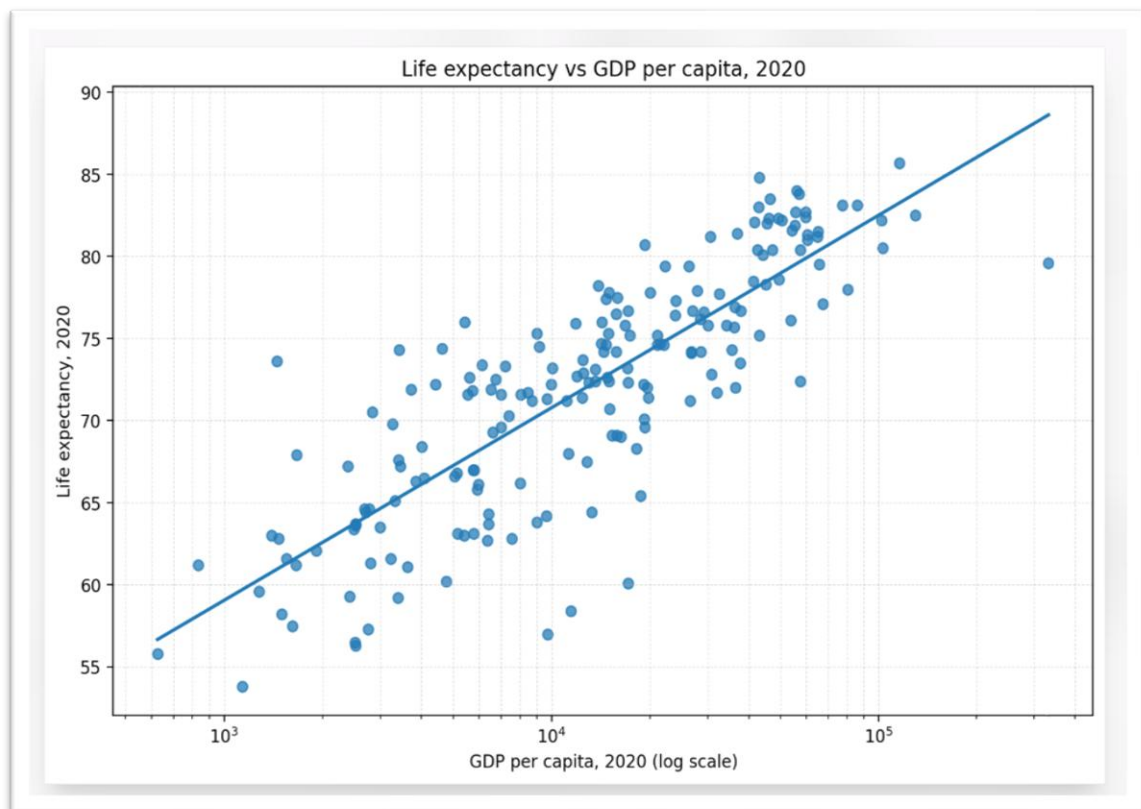


Fig. 4: Scatterplot – Life Expectancy vs GDP per capita, 2020



Brief interpretation

The scatterplot shows a **clear positive relationship** between GDP per capita and life expectancy in 2020. Countries with higher GDP per capita generally have higher life expectancy. The relationship is stronger when GDP per capita is viewed on a log scale, which is appropriate because GDP per capita is highly skewed.

Association measures from the merged 2020 data:

Measure	Value
Pearson correlation, raw GDP per capita	0.575
Pearson correlation, log GDP per capita	0.829
Spearman rank correlation	0.840

This suggests that increases in GDP per capita are strongly associated with higher life expectancy, especially at lower- and middle-income levels. At higher GDP levels, the gains in life expectancy appear to level off, which is why the log-scale relationship is more informative than the raw-scale relationship

Reflection:

I learnt a lot from this activity.

ChatGPT was the AI tool used for this assignment, which made it easy to analyse large data sets given, provided they were complete and readily available. It was very time efficient and to have a visual representation almost immediately was impressive. It was therefore easy to compile the report once all the plots were downloaded.

It was a bit difficult at first to give clear prompts of what you wanted especially to differentiate with different data such as the GDP per capita map above. Initially, the map provided showed similar colours and therefore hard to determine which category each country fit. After editing the prompt and specifying to have different colours to easily differentiate between the countries, the information became clearer hence Fig 3.

It was surprising how quickly the computational algorithm took with such a large data set and narrowing it down to the specific and requested year. This process definitely saved a lot of time and will be very useful to use in large research projects for work.